

# Supplementary Information

## Permitless Concealed Carry Adoption and Firearm Mortality in U.S. States, 1999–2024

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### Supplementary Note 1. Evidence hierarchy

The main manuscript interprets results through a hierarchy rather than treating every model row as an independent headline finding. Firearm suicide is the primary outcome. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths are secondary outcome-family checks. Cancer, cardiovascular, motor-vehicle, fall, other non-transport injury excluding falls and accidental poisoning, and accidental-poisoning mortality are negative-control or despair-environment diagnostics. Firearm-law covariates, contextual covariate layers, age-adjusted outcome rates, fractional-year treatment timing, covariate-balanced TWFE, Arkansas recoding, leave-one-out influence, and placebo timing are sensitivity checks. Cohort-window ATT, not-yet-treated ATT, and stacked DiD rows are alternative staggered-adoption estimands; event-time rows are timing diagnostics because they test pre-adoption signal. Rurality, gun ownership, baseline-risk heterogeneity, sex/age stratification, policy-feature interactions, and NICS mechanism proxies are exploratory or supplemental.

| Tier                         | Items                                                                                                                                                                                                                    | Interpretation                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Primary outcome              | Firearm suicide                                                                                                                                                                                                          | Central mortality claim.                                  |
| Secondary outcomes           | Total suicide; non-firearm suicide; firearm homicide; total firearm deaths                                                                                                                                               | Specificity and outcome-family interpretation.            |
| Sensitivity checks           | Firearm-law covariates; health-access and overdose covariates; Phase 3B2 covariate layers; age-adjusted outcome rates; fractional-year timing; covariate-balanced TWFE; Arkansas recoding; leave-one-out; placebo timing | Tests stability and boundaries.                           |
| Alternative timing estimands | Cohort-window ATT, not-yet-treated ATT summaries, and stacked DiD                                                                                                                                                        | Tests whether the positive association is unique to TWFE. |
| Exploratory checks           | Rurality, gun-ownership, baseline-risk, policy-feature heterogeneity, and NICS mechanism proxies                                                                                                                         | Hypothesis-generating or supplemental only.               |
| Appendix material            | Event-time rows; legal source table; full robustness grids                                                                                                                                                               | Transparency and audit trail.                             |

### Supplementary Note 2. Legal audit and policy-feature coding

The legal audit records one row per state and distinguishes clean source-verified within-panel adopters from verified non-adopters, partial cases, ambiguous cases, and baseline-permitless states. Partial, ambiguous, and baseline-permitless rows are documented so that they are not silently absorbed into the binary treatment definition.

| Audit status                    | State count |
|---------------------------------|-------------|
| Source-verified current adopter | 26          |
| Verified non-adopter            | 21          |
| Ambiguous reviewed              | 1           |
| Partial                         | 1           |
| Baseline permitless             | 1           |

Among the 26 clean source-verified adopter rows, 21 had a training requirement removed, 25 removed the carry-permit background-check screen, and 12 removed a permit-specific misdemeanor-violence screen. These policy-feature fields describe the carry-permit process and are treated as exploratory.

| Mechanism field/value                                       | State count |
|-------------------------------------------------------------|-------------|
| Training requirement removed: yes                           | 21          |
| Training requirement removed: no prior training requirement | 5           |
| Carry-permit background-check screen removed                | 25          |
| Carry-permit screen removed while purchase permit retained  | 1           |
| Permit-specific misdemeanor screen removed                  | 12          |
| Eligibility standard retained but no pre-carry check        | 7           |
| Other misdemeanor-screen coding categories                  | 7           |

The full state-level audit below reports the adoption year, audit status, whether the state enters the primary processed treatment map, the coding note, and source URLs used for legal verification. This table is included to make Arkansas, Mississippi, and Vermont visible rather than hiding the edge cases inside a binary treatment variable.

Table 1: State-by-state permitless concealed-carry treatment coding audit.

| State       | Year | Audit status         | Primary map | Coding note                                                                              | Source URL(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------|----------------------|-------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alabama     | 2023 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                                              | <a href="https://legiscan.com/AL/text/HB272/id/2544995">https://legiscan.com/AL/text/HB272/id/2544995</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Alaska      | 2003 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                                              | <a href="https://dps.alaska.gov/Statewide/R-I/PermitsLicensing/Firearm-FAQ">https://dps.alaska.gov/Statewide/R-I/PermitsLicensing/Firearm-FAQ</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Arizona     | 2010 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                                              | <a href="https://www.azleg.gov/legtext/49leg/2r/bills/sb1108s">https://www.azleg.gov/legtext/49leg/2r/bills/sb1108s</a> .<br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Arkansas    | –    | ambiguous reviewed   | No          | Excluded from primary clean-adoption map; 2021 and 2023 codings tested as sensitivities. | <a href="https://arkleg.state.ar.us/Bills/Detail?ddBienniumSess">https://arkleg.state.ar.us/Bills/Detail?ddBienniumSess</a><br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| California  | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.                           | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/</a> ;<br><a href="https://oag.ca.gov/firearms/ccw">https://oag.ca.gov/firearms/ccw</a><br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://cbi.colorado.gov/firearms/concealed-handgun-permit-chp-reciprocity">https://cbi.colorado.gov/firearms/concealed-handgun-permit-chp-reciprocity</a> |
| Colorado    | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.                           | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://cbi.colorado.gov/firearms/concealed-handgun-permit-chp-reciprocity">https://cbi.colorado.gov/firearms/concealed-handgun-permit-chp-reciprocity</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Connecticut | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.                           | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/</a> ;<br><a href="https://portal.ct.gov/despp/division-of-state-police/special-licensing-and-firearms/pistol-permits">https://portal.ct.gov/despp/division-of-state-police/special-licensing-and-firearms/pistol-permits</a>                                                                                                                                                                                                                                                                                                                                                                                                                          |

| State         | Year | Audit status         | Primary map | Coding note                                                    | Source URL(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------|------|----------------------|-------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Delaware      | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024. | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://attorneygeneral.delaware.gov/criminal/concealed-carry-deadly-weapons-ccdwl/">https://attorneygeneral.delaware.gov/criminal/concealed-carry-deadly-weapons-ccdwl/</a> |
| Florida       | 2023 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                    | <a href="https://www.flgov.com/eog/news/press/2023/governor-ron-desantis-signs-hb-543-constitutional-carry">https://www.flgov.com/eog/news/press/2023/governor-ron-desantis-signs-hb-543-constitutional-carry</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Carry-in-Public'and'Concealed-Carry'">https://www.handgunlaw.us/documents/Permitless'Carry-in-Public'and'Concealed-Carry'</a>                                                                                                                                                                  |
| Georgia       | 2022 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                    | <a href="https://gov.georgia.gov/press-releases/2022-04-13/gov-kemp-signs-georgia-constitutional-carry-act-law">https://gov.georgia.gov/press-releases/2022-04-13/gov-kemp-signs-georgia-constitutional-carry-act-law</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Carry-in-Public'and'Concealed-Carry'">https://www.handgunlaw.us/documents/Permitless'Carry-in-Public'and'Concealed-Carry'</a>                                                                                                                                                          |
| Hawaii        | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024. | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/</a> ;<br><a href="https://www.honolulupd.org/police-services/firearms/">https://www.honolulupd.org/police-services/firearms/</a>                                                                       |
| Idaho         | 2016 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                    | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Carry-in-Public'and'Concealed-Carry'">https://www.handgunlaw.us/documents/Permitless'Carry-in-Public'and'Concealed-Carry'</a>                                                                                                                                                                                            |
| Illinois      | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024. | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/</a> ;<br><a href="https://isp.illinois.gov/Foid/Ccl">https://isp.illinois.gov/Foid/Ccl</a>                                                                                                             |
| Indiana       | 2022 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                    | <a href="https://events.in.gov/event/gov%27holcomb%27issues%27statement">https://events.in.gov/event/gov%27holcomb%27issues%27statement</a> ;<br><a href="https://www.in.gov/isp/files/Permitless-Carry-Website-Messaging.pdf">https://www.in.gov/isp/files/Permitless-Carry-Website-Messaging.pdf</a>                                                                                                                                                                                                                                                                        |
| Iowa          | 2021 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                    | <a href="https://www.legis.iowa.gov/legislation/billTracking/bills/2021/2021%2016/b2015%2016/measures">https://www.legis.iowa.gov/legislation/billTracking/bills/2021/2021%2016/b2015%2016/measures</a>                                                                                                                                                                                                                                                                                                                                                                       |
| Kansas        | 2015 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                    | <a href="https://www.kslegislature.gov/li/2016/b2015%2016/measures">https://www.kslegislature.gov/li/2016/b2015%2016/measures</a>                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Kentucky      | 2019 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                    | <a href="https://www.handgunlaw.us/documents/Permitless'Carry-in-Public'and'Concealed-Carry'">https://www.handgunlaw.us/documents/Permitless'Carry-in-Public'and'Concealed-Carry'</a> ;<br><a href="https://www.kentuckystatepolice.ky.gov/ccdw-home/">https://www.kentuckystatepolice.ky.gov/ccdw-home/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a>                                                                     |
| Louisiana     | 2024 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                    | <a href="https://legis.la.gov/Legis/BillInfo.aspx?i=245605">https://legis.la.gov/Legis/BillInfo.aspx?i=245605</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a>                                                                                                                                                                                                                                                                |
| Maine         | 2015 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                    | <a href="https://www.maine.gov/dps/msp/licenses-permits/concealed-carry-maine">https://www.maine.gov/dps/msp/licenses-permits/concealed-carry-maine</a> ;<br><a href="https://legislature.maine.gov/legis/bills/display'ps.aspx">https://legislature.maine.gov/legis/bills/display'ps.aspx</a>                                                                                                                                                                                                                                                                                |
| Maryland      | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024. | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/</a> ;<br><a href="https://mdsp.maryland.gov/Organization/Pages/Criminal">https://mdsp.maryland.gov/Organization/Pages/Criminal</a>                                                                     |
| Massachusetts | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024. | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/</a> ;<br><a href="https://www.mass.gov/how-to/apply-for-a-firearms-license">https://www.mass.gov/how-to/apply-for-a-firearms-license</a>                                                               |
| Michigan      | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024. | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://www.michigan.gov/msp/services/ccw">https://www.michigan.gov/msp/services/ccw</a>                                                                                     |

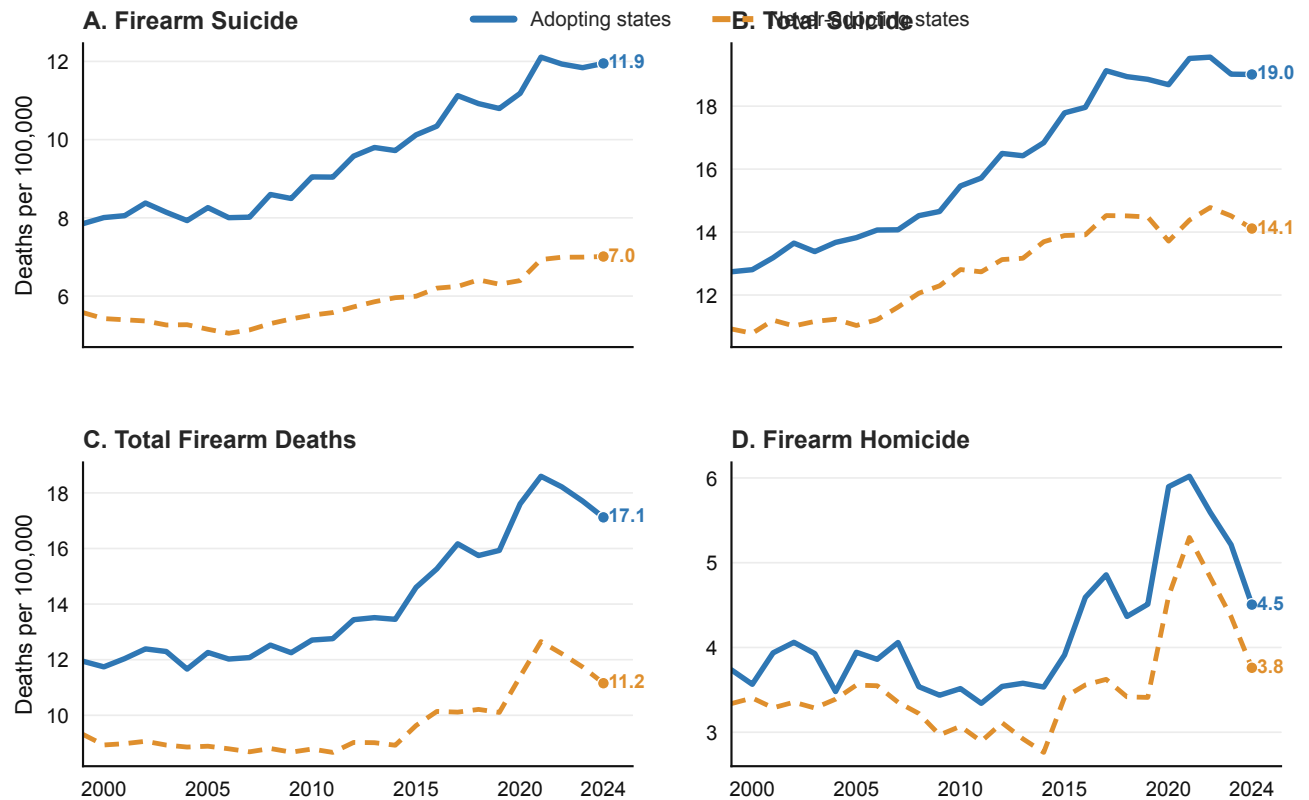
| State          | Year | Audit status         | Primary map | Coding note                                                                      | Source URL(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| Minnesota      | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.                   | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/</a> ;<br><a href="https://dps.mn.gov/divisions/bca/bca-divisions/administrative/Pages/Permit-to-Carry.aspx">https://dps.mn.gov/divisions/bca/bca-divisions/administrative/Pages/Permit-to-Carry.aspx</a><br><a href="https://billstatus.ls.state.ms.us/documents/2016/html/0799/HB0786PS.htm">https://billstatus.ls.state.ms.us/documents/2016/html/0799/HB0786PS.htm</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a><br><a href="https://www.senate.mo.gov/BillTracking/Bills/BillInfo">https://www.senate.mo.gov/BillTracking/Bills/BillInfo</a><br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a><br><a href="https://legiscan.com/MT/bill/HB102/2021">https://legiscan.com/MT/bill/HB102/2021</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a><br><a href="https://nebraskalegislature.gov/bills/view'bill.php?Doc">https://nebraskalegislature.gov/bills/view'bill.php?Doc</a><br><a href="https://statepatrol.nebraska.gov/services/concealed-handgun-permits/chp-frequently-asked-questions">https://statepatrol.nebraska.gov/services/concealed-handgun-permits/chp-frequently-asked-questions</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Mississippi    | 2015 | partial              | Yes         | Retained in processed panel as 2015 adoption; legal audit marks the row partial. | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://rccd.nv.gov/FeesForms/Brady/CCW-Permit-Recognition/">https://rccd.nv.gov/FeesForms/Brady/CCW-Permit-Recognition/</a><br><a href="https://legiscan.com/NH/votes/SB12/2017">https://legiscan.com/NH/votes/SB12/2017</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Missouri       | 2017 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                                      | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/</a> ;<br><a href="https://www.nj.gov/njsp/firearms/forms.shtml">https://www.nj.gov/njsp/firearms/forms.shtml</a><br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://www.dps.nm.gov/concealed-carry/">https://www.dps.nm.gov/concealed-carry/</a><br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/</a> ;<br><a href="https://gunsafety.ny.gov/pistol-permit-recertification">https://gunsafety.ny.gov/pistol-permit-recertification</a><br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://ncdoj.gov/law-enforcement-training/criminal-justice/officer-certification-programs/concealed-weapon-reciprocity/">https://ncdoj.gov/law-enforcement-training/criminal-justice/officer-certification-programs/concealed-weapon-reciprocity/</a><br><a href="https://www.nd.gov/ndhp/node/116">https://www.nd.gov/ndhp/node/116</a> ;<br><a href="https://legiscan.com/ND/bill/HB1169/2017">https://legiscan.com/ND/bill/HB1169/2017</a><br><a href="https://www.legislature.ohio.gov/download?key=1896">https://www.legislature.ohio.gov/download?key=1896</a><br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a><br><a href="https://www.oklegislature.gov/cf/pdf/2019-20%20SUPPORT%20DOCUMENTS/BILLSUM/Hou">https://www.oklegislature.gov/cf/pdf/2019-20%20SUPPORT%20DOCUMENTS/BILLSUM/Hou</a><br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a><br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://www.oregon.gov/osp/programs/cjis/pages/oregon-concealed-handgun-license.aspx">https://www.oregon.gov/osp/programs/cjis/pages/oregon-concealed-handgun-license.aspx</a> |
| Montana        | 2021 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Nebraska       | 2023 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Nevada         | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| New Hampshire  | 2017 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| New Jersey     | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| New Mexico     | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| New York       | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| North Carolina | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| North Dakota   | 2017 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Ohio           | 2022 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Oklahoma       | 2019 | clean adopter        | Yes         | Clean source-verified within-panel adopter.                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Oregon         | –    | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| State          | Year      | Audit status         | Primary map | Coding note                                                              | Source URL(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|-----------|----------------------|-------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pennsylvania   | –         | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.           | <a href="https://giffords.org/lawcenter/state-laws/concealed-carry-in-pennsylvania/">https://giffords.org/lawcenter/state-laws/concealed-carry-in-pennsylvania/</a> ;<br><a href="https://giffords.org/lawcenter/state-laws/open-carry-in-pennsylvania/">https://giffords.org/lawcenter/state-laws/open-carry-in-pennsylvania/</a> ;<br><a href="https://www.pa.gov/en/agencies/psp/programs/firearms-in-pennsylvania.html">https://www.pa.gov/en/agencies/psp/programs/firearms-in-pennsylvania.html</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Rhode Island   | –         | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.           | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/</a> ;<br><a href="https://riag.ri.gov/i-want/get-concealed-carry-permit">https://riag.ri.gov/i-want/get-concealed-carry-permit</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| South Carolina | 2024      | clean adopter        | Yes         | Clean source-verified within-panel adopter.                              | <a href="https://sled.sc.gov/constitutional-carry-guidance">https://sled.sc.gov/constitutional-carry-guidance</a> ;<br><a href="https://legiscan.com/SC/supplement/H3594/id/47310">https://legiscan.com/SC/supplement/H3594/id/47310</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| South Dakota   | 2019      | clean adopter        | Yes         | Clean source-verified within-panel adopter.                              | <a href="https://sdlegislature.gov/Session/Bill/9374">https://sdlegislature.gov/Session/Bill/9374</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Tennessee      | 2021      | clean adopter        | Yes         | Clean source-verified within-panel adopter.                              | <a href="https://www.wvltv.com/2021/04/08/governor-lee-signs-permitless-carry-into-law/">https://www.wvltv.com/2021/04/08/governor-lee-signs-permitless-carry-into-law/</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Texas          | 2021      | clean adopter        | Yes         | Clean source-verified within-panel adopter.                              | <a href="https://www.lrl.texas.gov/legis/billsearch/billdetails.cfm">https://www.lrl.texas.gov/legis/billsearch/billdetails.cfm</a> ;<br><a href="https://sll.texas.gov/faqs/permitless-carry-guns/">https://sll.texas.gov/faqs/permitless-carry-guns/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Utah           | 2021      | clean adopter        | Yes         | Clean source-verified within-panel adopter.                              | <a href="https://bci.utah.gov/concealed-firearm/">https://bci.utah.gov/concealed-firearm/</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Vermont        | pre-panel | baseline permitless  | No          | Baseline permitless before the panel; not a within-panel adoption event. | <a href="https://www.nraila.org/gun-laws/state-gun-laws/vermont/">https://www.nraila.org/gun-laws/state-gun-laws/vermont/</a> ;<br><a href="https://www.handgunlaw.us/states/vermont.pdf">https://www.handgunlaw.us/states/vermont.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Virginia       | –         | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.           | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://vsp.virginia.gov/services/firearms/resident-concealed-handgun-permits/">https://vsp.virginia.gov/services/firearms/resident-concealed-handgun-permits/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://dof.wa.gov/professional-licenses/concealed-pistol-license">https://dof.wa.gov/professional-licenses/concealed-pistol-license</a> |
| Washington     | –         | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.           | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://dof.wa.gov/professional-licenses/concealed-pistol-license">https://dof.wa.gov/professional-licenses/concealed-pistol-license</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| West Virginia  | 2016      | clean adopter        | Yes         | Clean source-verified within-panel adopter.                              | <a href="https://wvpress.org/breaking-news/wvs-no-permit-concealed-carry-begins-today/">https://wvpress.org/breaking-news/wvs-no-permit-concealed-carry-begins-today/</a> ;<br><a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Wisconsin      | –         | verified non-adopter | No          | No statewide permitless concealed-carry adoption through 2024.           | <a href="https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/">https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/</a> ;<br><a href="https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/">https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/</a> ;<br><a href="https://www.wisdoj.gov/Pages/PublicSafety/concealed-carry-weapon-license.aspx">https://www.wisdoj.gov/Pages/PublicSafety/concealed-carry-weapon-license.aspx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Wyoming        | 2011      | clean adopter        | Yes         | Clean source-verified within-panel adopter.                              | <a href="https://legiscan.com/WY/bill/SF0047/2011">https://legiscan.com/WY/bill/SF0047/2011</a> ;<br><a href="https://www.handgunlaw.us/documents/Permitless'Ca">https://www.handgunlaw.us/documents/Permitless'Ca</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

### Supplementary Note 3. Main and secondary mortality estimates

| Outcome              | Coef.  | SE    | <i>p</i> | N    |
|----------------------|--------|-------|----------|------|
| Firearm suicide      | 1.391  | 0.222 | < 0.001  | 1222 |
| Non-firearm suicide  | 0.415  | 0.139 | 0.003    | 1222 |
| Total suicide        | 1.805  | 0.294 | < 0.001  | 1222 |
| Firearm homicide     | -0.234 | 0.338 | 0.489    | 1142 |
| Total firearm deaths | 1.271  | 0.482 | 0.008    | 1222 |

Descriptive outcome trends are included as a supporting check, not as causal evidence. They show why the paper separates suicide-related outcomes from homicide outcomes before fitting adjusted models.



Annual state means, 1999-2024. Groups are defined by whether a state ever adopts permitless carry during the panel.

Figure 1: **Descriptive outcome trends by permitless-carry adoption status.** Firearm-suicide and total-suicide rates were higher in adopter states, while homicide trends were noisier and did not visually follow the same pattern. The figure is descriptive because adopter and non-adopter states differed before adoption.

## Supplementary Note 4. Firearm-suicide specification and timing summary

The summary places the baseline fixed-effects estimate beside population weighting, COVID-window restrictions, leave-one-adopter-out checks, placebo timing, state-specific trends, cohort-window ATT estimates, and event-time diagnostics. It is the main table for interpreting direction, magnitude sensitivity, and identification fragility.

Table 2: **Firearm-suicide specification and timing summary.**

| Diagnostic row                    | Estimate          | Diag. $p$ | Interpretation                                                     |
|-----------------------------------|-------------------|-----------|--------------------------------------------------------------------|
| Baseline unweighted fixed effects | 1.391             | $< 0.001$ | Clean-adoption state-level association                             |
| Population-weighted fixed effects | 0.919             | $< 0.001$ | Resident-weighted association                                      |
| Pre-COVID only                    | 1.298             | $< 0.001$ | Drops post-2019 years                                              |
| COVID-excluded                    | 1.356             | $< 0.001$ | Drops 2020–2021                                                    |
| External firearm-law covariates   | 1.096             | $< 0.001$ | Policy-environment sensitivity                                     |
| Leave-one-adopter-out             | 1.265–1.481       | –         | No sign changes across adopter exclusions                          |
| Never-treated placebo timing      | 1.391 $> 0.338$   | –         | Observed estimate exceeds placebo timing distribution              |
| State-specific linear trends      | 0.249             | 0.079     | Central attenuation caveat                                         |
| Cohort ATT windows                | 0.601/0.577/0.641 | –         | Alternative staggered-adoption estimand                            |
| Not-yet-treated dynamic ATT       | 0.387             | –         | Alternative timing comparison                                      |
| Event-time fixed effects          | 0.651             | 0.0000001 | Positive post mean; diagnostic $p$ is the minimum pre-adoption $p$ |

Estimates are deaths per 100,000 residents. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors unless otherwise noted. Cohort-window and not-yet-treated ATT rows are alternative timing estimands. The event-time row is a pretrend diagnostic; its  $p$  column reports the minimum pre-adoption  $p$  rather than a post-adoption effect test.

The focused robustness table below reports the estimates used in the main-text robustness paragraph: binary TWFE, fractional-year TWFE, stacked DiD, covariate-balanced TWFE, and firearm suicide minus non-firearm suicide.

Table 3: **Primary firearm-suicide robustness estimates.**

| Specification                          | Estimate | SE    | $p$       | N    | Interpretation                                                   |
|----------------------------------------|----------|-------|-----------|------|------------------------------------------------------------------|
| Binary TWFE                            | 1.391    | 0.222 | $< 0.001$ | 1222 | Headline annual treatment coding                                 |
| Fractional-year TWFE                   | 1.534    | 0.239 | $< 0.001$ | 1222 | Effective-year exposure prorated by law date                     |
| Stacked DiD                            | 0.665    | 0.137 | $< 0.001$ | 3446 | Cohort stacks with not-yet-treated controls                      |
| Covariate-balanced TWFE                | 0.889    | 0.304 | 0.003     | 1222 | Non-adopters reweighted toward adopter covariates                |
| Firearm minus non-firearm suicide TWFE | 0.976    | 0.226 | $< 0.001$ | 1222 | Firearm-specific association net of non-firearm suicide movement |

Estimates are deaths per 100,000 residents. The first four rows estimate firearm suicide. The final TWFE row estimates firearm suicide minus non-firearm suicide, so it is a firearm-specific check rather than a total-suicide effect. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. The stacked DiD row uses 5-year pre/post cohort stacks with not-yet-treated controls.

A null-imposed Rademacher wild-cluster bootstrap with 999 replications over 47 state clusters yielded  $p = 0.001$  for the baseline firearm-suicide row. The same bootstrap gave  $p = 0.016$  for total firearm deaths and  $p = 0.509$  for firearm homicide. These values address small-cluster inference for the baseline TWFE rows; they do not address the separate pretrend, trend-specification, or common-support limitations.

Total firearm deaths are retained as a secondary outcome-family check rather than a headline finding. The baseline total-firearm estimate was positive (1.271;  $p = 0.008$ ), but it attenuated to 0.070 under state-specific trends ( $p = 0.806$ ) and 0.141 under covariate balancing ( $p = 0.851$ ).

## Supplementary Note 5. Age-adjusted outcome-rate sensitivity

Crude mortality rates are the primary outcome scale because they measure realized state-year mortality burden. Age-adjusted rates are reported as a demographic-composition sensitivity check using CDC WONDER age-adjusted

rates standardized to the 2000 U.S. standard population. The age-adjusted estimates preserve the crude-rate pattern: firearm suicide, total suicide, and total firearm deaths remain positive, while firearm homicide remains statistically weak and centered below zero. The total-firearm row is not interpreted as a stable headline result because later trend and covariate-balancing checks attenuate it.

| Outcome                            | Coef.  | SE    | <i>p</i> | N    |
|------------------------------------|--------|-------|----------|------|
| Firearm suicide, age-adjusted      | 1.407  | 0.228 | < 0.001  | 1220 |
| Total suicide, age-adjusted        | 1.914  | 0.327 | < 0.001  | 1222 |
| Total firearm deaths, age-adjusted | 1.230  | 0.510 | 0.016    | 1222 |
| Firearm homicide, age-adjusted     | -0.230 | 0.426 | 0.590    | 1037 |

## Supplementary Note 6. Firearm-law and contextual covariate sensitivity

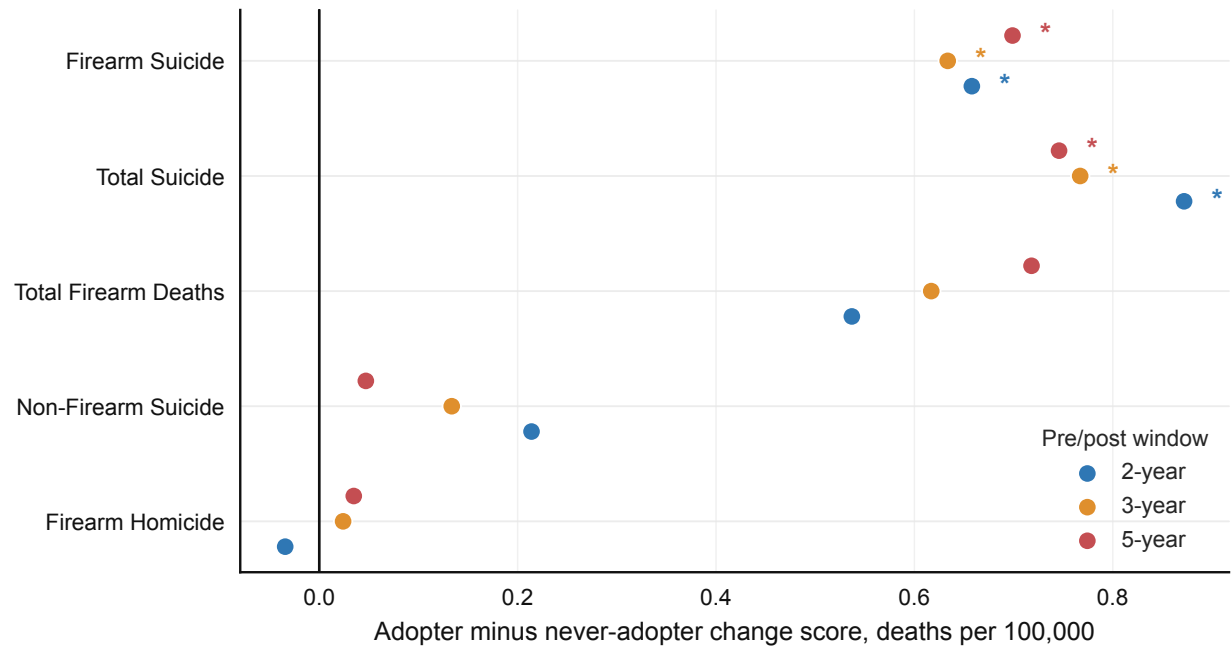
External firearm-law covariates, health-access covariates, and demographic-poverty covariates attenuated but did not erase the firearm-suicide estimate. Overdose controls reduced precision in a short sample. The short HRSA mental-health-provider window did not retain statistical significance and should be interpreted as a short-window sensitivity rather than a full-panel replacement. These rows are covariate-layer sensitivity checks, not proof that adding more time-varying covariates necessarily improves identification.

| Firearm-suicide specification            | Coef. | <i>p</i> | N    |
|------------------------------------------|-------|----------|------|
| Baseline fixed effects                   | 1.391 | < 0.001  | 1222 |
| External firearm-law covariates          | 1.096 | < 0.001  | 1222 |
| Health-access covariates                 | 0.707 | < 0.001  | 752  |
| Overdose covariates                      | 0.492 | 0.060    | 282  |
| Demographic-poverty covariates           | 0.695 | < 0.001  | 940  |
| Mental-health-provider access covariates | 0.116 | 0.744    | 141  |
| Full Phase 3B2 covariates                | 0.176 | 0.633    | 141  |

## Supplementary Note 7. Change-score checks

| Outcome              | 2-year diff. | 3-year diff. | 5-year diff. |
|----------------------|--------------|--------------|--------------|
| Firearm suicide      | 0.658        | 0.634        | 0.699        |
| Non-firearm suicide  | 0.214        | 0.134        | 0.047        |
| Total suicide        | 0.872        | 0.767        | 0.746        |
| Firearm homicide     | -0.034       | 0.024        | 0.035        |
| Total firearm deaths | 0.537        | 0.617        | 0.718        |





Positive values indicate larger pre/post increases in adopting states. Asterisks denote Welch-test  $p < 0.05$ .

Figure 2: **Change-score robustness across pre-post windows.** Firearm suicide is positive across 2-, 3-, and 5-year windows. Firearm homicide remains statistically weak across the same windows. These window checks support the suicide-homicide contrast but are less adjusted than the fixed-effects models.

## Supplementary Note 8. Staggered-adoption estimates and event-time diagnostics

| Outcome              | ATT w2 | ATT w3 | ATT w5 | Not-yet post | Interpretation                |
|----------------------|--------|--------|--------|--------------|-------------------------------|
| Firearm suicide      | 0.601  | 0.577  | 0.641  | 0.387        | Positive, smaller than TWFE.  |
| Non-firearm suicide  | 0.270  | 0.239  | 0.241  | 0.077        | Positive, smaller than TWFE.  |
| Total suicide        | 0.871  | 0.816  | 0.882  | 0.464        | Positive, with pretrend flag. |
| Firearm homicide     | -0.004 | -0.107 | -0.106 | -0.157       | No positive estimate.         |
| Total firearm deaths | 0.382  | 0.435  | 0.493  | 0.200        | Positive, with pretrend flag. |

The stacked DiD specification estimated a firearm-suicide association of 0.665 deaths per 100,000 (SE 0.137;  $p < 0.001$ ). This estimate is smaller than the baseline binary TWFE estimate and closer to the cohort-window ATT rows, which supports the manuscript's interpretation that the direction is more stable than the exact magnitude.

## Supplementary Note 9. Robustness and Arkansas sensitivity

| Outcome              | Baseline | P05 specs | Leave-one min | Leave-one max | Flag                              |
|----------------------|----------|-----------|---------------|---------------|-----------------------------------|
| Firearm suicide      | 1.391    | 5         | 1.265         | 1.481         | Stable positive, trend-sensitive. |
| Non-firearm suicide  | 0.415    | 5         | 0.371         | 0.472         | Stable positive, trend-sensitive. |
| Total suicide        | 1.805    | 6         | 1.656         | 1.954         | Stable positive.                  |
| Firearm homicide     | -0.234   | 0         | -0.399        | -0.049        | No stable positive signal.        |
| Total firearm deaths | 1.271    | 4         | 1.058         | 1.532         | Positive, trend-sensitive.        |

Arkansas recoding as 2021 or 2023 retained the sign of all five outcomes. Firearm suicide, total suicide, non-firearm suicide, and total firearm deaths retained  $p < 0.05$  across both Arkansas codings; firearm homicide did not. Reintroducing Mississippi alone and in combination with both Arkansas codings preserved the positive firearm-suicide and total-suicide pattern.

Table 4: **Firearm-suicide leave-one-adopter-out estimates by excluded state.**

| Excluded state | Coef. | $p$     |
|----------------|-------|---------|
| Alabama        | 1.415 | < 0.001 |
| Alaska         | 1.405 | < 0.001 |
| Arizona        | 1.481 | < 0.001 |
| Florida        | 1.398 | < 0.001 |
| Georgia        | 1.413 | < 0.001 |
| Idaho          | 1.375 | < 0.001 |
| Indiana        | 1.393 | < 0.001 |
| Iowa           | 1.395 | < 0.001 |
| Kansas         | 1.414 | < 0.001 |
| Kentucky       | 1.451 | < 0.001 |
| Louisiana      | 1.411 | < 0.001 |
| Maine          | 1.399 | < 0.001 |
| Missouri       | 1.366 | < 0.001 |
| Montana        | 1.342 | < 0.001 |
| Nebraska       | 1.404 | < 0.001 |
| New Hampshire  | 1.391 | < 0.001 |
| North Dakota   | 1.358 | < 0.001 |
| Ohio           | 1.410 | < 0.001 |
| Oklahoma       | 1.383 | < 0.001 |
| South Carolina | 1.406 | < 0.001 |
| South Dakota   | 1.403 | < 0.001 |
| Tennessee      | 1.416 | < 0.001 |
| Texas          | 1.415 | < 0.001 |
| Utah           | 1.424 | < 0.001 |
| West Virginia  | 1.420 | < 0.001 |
| Wyoming        | 1.265 | < 0.001 |

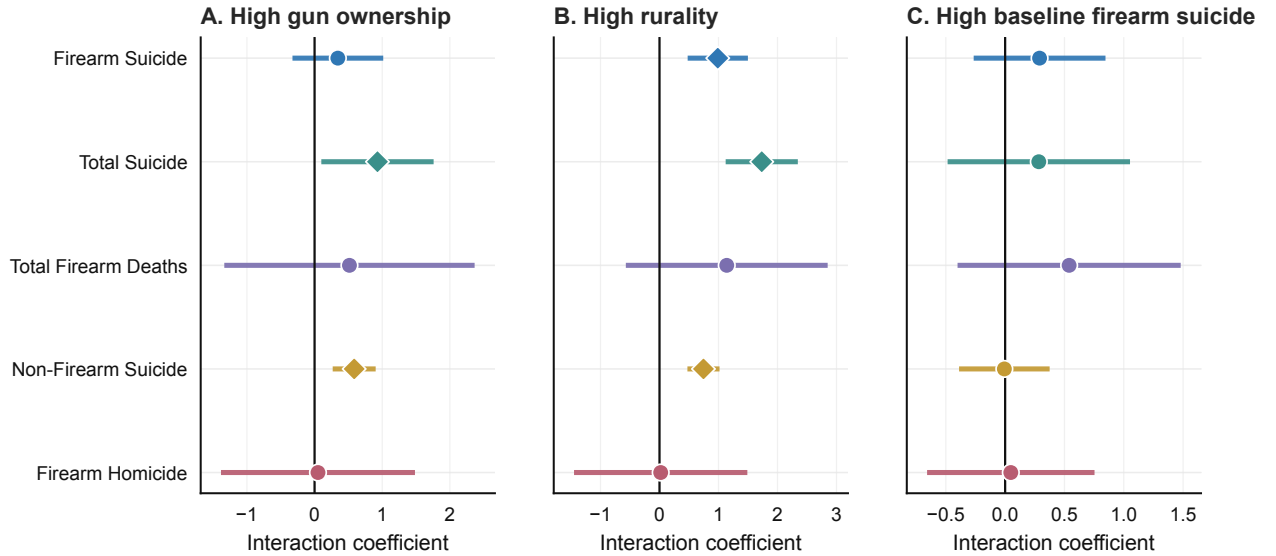
Table 5: **Firearm-suicide never-treated placebo timing estimates.**

| Placebo shift | Coef.  | <i>p</i> |
|---------------|--------|----------|
| 0             | -0.026 | 0.850    |
| 1             | -0.246 | 0.173    |
| 2             | -0.062 | 0.788    |
| 3             | 0.081  | 0.721    |
| 4             | 0.385  | 0.060    |
| 5             | 0.291  | 0.047    |
| 6             | 0.054  | 0.675    |
| 7             | 0.050  | 0.767    |
| 8             | -0.032 | 0.883    |
| 9             | -0.224 | 0.375    |
| 10            | -0.275 | 0.156    |

The absolute median placebo estimate was 0.081 deaths per 100,000 and the absolute 95th percentile was 0.338. The observed firearm-suicide estimate was 1.391, exceeding that placebo timing distribution.

## Supplementary Note 10. Heterogeneity and policy-feature checks

Rurality was the clearest recurring heterogeneity dimension: firearm suicide, non-firearm suicide, and total suicide had positive high-rurality interactions. These models are exploratory because state-level subgroup indicators are coarse and adoption is politically patterned.



Interaction terms estimate whether associations are larger in states above each baseline median; diamonds mark  $p < 0.05$ .

Figure 3: **Exploratory heterogeneity checks.** Rurality is the clearest recurring subgroup pattern for suicide-related outcomes. These results describe where the association was larger; they do not identify the mechanism.

| Outcome/dimension                       | Interaction | <i>p</i> |
|-----------------------------------------|-------------|----------|
| Firearm suicide, high rurality          | 0.988       | < 0.001  |
| Total suicide, high rurality            | 1.733       | < 0.001  |
| Non-firearm suicide, high rurality      | 0.745       | < 0.001  |
| Non-firearm suicide, high gun ownership | 0.586       | < 0.001  |

Policy-feature heterogeneity did not identify a stable mechanism. None of the 15 training, carry-permit background-check, or misdemeanor-screen interaction rows had  $p < 0.05$ . Five rows were sparse or unavailable

because the feature-specific clean-adopter comparison group was absent.

## Supplementary Note 11. NICS mechanism proxy and additional CDC WONDER checks

The NICS mechanism proxy uses the official FBI Year by State/Type table where local text extraction produced reliable state-year rows. The processed panel covers 50 states from 1999 through 2021. The direct handgun-check proxy increased after permitless-carry adoption, but the permit and combined handgun-or-permit rows do not show a net increase. These rows are supplemental because NICS checks are not firearm sales and because the locally parsed official file did not produce clean 2022–2024 state-year rows.

| NICS proxy                             | Sample              | Coef.     | SE       | <i>p</i> |
|----------------------------------------|---------------------|-----------|----------|----------|
| Handgun checks per 100,000             | 1999–2021           | 476.952   | 185.444  | 0.010    |
| Permit checks per 100,000              | 1999–2021           | -2274.542 | 1846.033 | 0.218    |
| Permit plus recheck checks per 100,000 | 1999–2021           | -643.925  | 1240.223 | 0.604    |
| Handgun or permit checks per 100,000   | 1999–2021           | -1797.590 | 1874.740 | 0.338    |
| Handgun checks per 100,000             | Excluding 2020–2021 | 379.432   | 175.813  | 0.031    |
| Handgun or permit checks per 100,000   | Excluding 2020–2021 | -976.463  | 1183.758 | 0.409    |

The negative-control mortality checks do not show a uniform positive mortality shift. Cancer mortality was positive but statistically weak, cardiovascular mortality was near zero, and motor-vehicle mortality moved in the opposite direction. The added injury and poisoning checks were mixed: fall mortality was positive but statistically weak, other non-transport injury excluding falls and accidental poisoning was small and negative, and accidental-poisoning mortality was negative and imprecise. The other non-transport row excludes both falls and accidental poisoning so the injury rows are not mechanically overlapping; the accidental-poisoning row directly tests the overdose/despair pathway. This does not make adopter and non-adopter states exchangeable, but it weakens the interpretation that the firearm-suicide estimate is merely a uniform rise across all mortality outcomes.

Table 6: **Negative-control mortality estimates.**

| Outcome                  | Coef.  | SE    | <i>p</i> | N    |
|--------------------------|--------|-------|----------|------|
| Cancer mortality         | 1.796  | 2.332 | 0.441    | 1222 |
| Cardiovascular mortality | -0.591 | 5.810 | 0.919    | 1222 |
| Motor-vehicle mortality  | -1.319 | 0.416 | 0.002    | 1222 |

Table 7: **Additional injury and accidental-poisoning mortality estimates.**

| Outcome                                              | Coef.  | SE    | <i>p</i> | N    |
|------------------------------------------------------|--------|-------|----------|------|
| Fall mortality                                       | 1.178  | 0.913 | 0.197    | 1222 |
| Other non-transport injury excluding falls/poisoning | -0.218 | 0.285 | 0.445    | 1222 |
| Accidental poisoning mortality                       | -1.484 | 1.943 | 0.445    | 1215 |

Sex/age stratification shows that the firearm-suicide association is concentrated among males. Female firearm-suicide estimates were small and statistically weak, while male estimates were positive overall and across broad adult age groups. These rows are state-level stratified outcomes, not individual-level mechanism evidence. The analysis uses complete broad-age cells after CDC WONDER suppression, so observation counts differ across strata.

Table 8: **Firearm-suicide estimates by sex and broad age group.**

| Stratum                    | Coef. | SE    | <i>p</i> | N    |
|----------------------------|-------|-------|----------|------|
| Female, modeled adult ages | 0.108 | 0.200 | 0.590    | 749  |
| Male, modeled adult ages   | 1.665 | 0.389 | < 0.001  | 1141 |
| Male, age 15-24            | 1.651 | 0.760 | 0.030    | 1085 |
| Male, age 25-44            | 1.702 | 0.483 | < 0.001  | 1038 |
| Male, age 45-64            | 1.830 | 0.495 | < 0.001  | 1066 |
| Male, age 65+              | 1.332 | 0.737 | 0.071    | 631  |

## Supplementary Note 12. Homicide missingness audit

Firearm-homicide outcome rows have fewer observations than suicide and total-firearm rows because some state-year rates are unavailable or suppressed. The missingness audit shows that all missing firearm-homicide rows occur in the smallest-population state-years.

Table 9: **Firearm-homicide missingness audit.**

| Group                         | Possible | Observed | Missing | Missing pct. |
|-------------------------------|----------|----------|---------|--------------|
| All state-years               | 1222     | 1142     | 80      | 6.5%         |
| Adopter-state state-years     | 676      | 610      | 66      | 9.8%         |
| Non-adopter-state state-years | 546      | 532      | 14      | 2.6%         |
| Q1 smallest                   | 312      | 232      | 80      | 25.6%        |
| Q2                            | 312      | 312      | 0       | 0.0%         |
| Q3                            | 286      | 286      | 0       | 0.0%         |
| Q4 largest                    | 312      | 312      | 0       | 0.0%         |

Missing firearm-homicide rates are concentrated in the smallest-population state-years. No state-year in population quartiles 2–4 is missing the firearm-homicide rate.

| State         | Missing | Possible | Missing pct. |
|---------------|---------|----------|--------------|
| New Hampshire | 16      | 26       | 61.5%        |
| North Dakota  | 15      | 26       | 57.7%        |
| South Dakota  | 14      | 26       | 53.8%        |
| Wyoming       | 14      | 26       | 53.8%        |
| Hawaii        | 13      | 26       | 50.0%        |
| Maine         | 6       | 26       | 23.1%        |
| Delaware      | 1       | 26       | 3.8%         |
| Montana       | 1       | 26       | 3.8%         |

## Supplementary Note 13. Full diagnostic source files

The following repository outputs provide the full machine-readable diagnostic trail used to write the manuscript and this supplement:

- `outputs/tables/did/twfe.did.main.results.csv`
- `outputs/tables/did/twfe.did.ageadj.sensitivity.summary.csv`
- `outputs/tables/did/twfe.did.firearm.law.control.summary.csv`
- `outputs/tables/did/twfe.did.nonfirearm.confounder.summary.csv`
- `outputs/tables/did/twfe.did.phase3b2.confounder.summary.csv`
- `outputs/tables/main/welch.change.score.results.csv`

- outputs/tables/modern\_did/modern\_did\_summary.csv
- outputs/tables/modern\_did/stacked\_did\_results.csv
- outputs/tables/modern\_did/event\_time\_never\_control\_estimates.csv
- outputs/tables/robustness/robustness\_summary.csv
- outputs/tables/robustness/fractional\_timing\_results.csv
- outputs/tables/robustness/covariate\_balanced\_twfe\_results.csv
- outputs/tables/robustness/covariate\_balance\_diagnostics.csv
- outputs/tables/robustness/firearm\_suicide\_main\_robustness\_table.csv
- outputs/tables/robustness/leave\_one\_state\_out.csv
- outputs/tables/robustness/placebo\_never\_treated.csv
- outputs/tables/robustness/arkansas\_treatment\_sensitivity\_summary.csv
- outputs/tables/heterogeneity/heterogeneity\_did\_results.csv
- outputs/tables/heterogeneity/firearm\_suicide\_sex\_age\_twfe\_results.csv
- outputs/tables/negative\_controls/negative\_control\_twfe\_results.csv
- outputs/tables/mechanism/mechanism\_heterogeneity\_results.csv
- outputs/tables/mechanism/nics\_handgun\_proxy\_results.csv
- outputs/tables/mechanism/policy\_bundle\_distribution.csv
- outputs/tables/data\_availability/additional\_analysis\_data\_availability.csv
- data/processed/analysis\_panel\_negative\_control\_mortality\_1999\_2024.csv
- data/processed/analysis\_panel\_firearm\_suicide\_by\_sex\_age\_1999\_2024.csv
- data/policy/permitless\_carry\_legal\_audit.csv